AlpsBench: An LLM Personalization Benchmark for Real-Dialogue Memorization and Preference Alignment

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Abstract

As Large Language Models (LLMs) evolve into lifelong AI assistants, LLM personalization has become a critical frontier. However, progress is currently bottlenecked by the absence of a gold-standard evaluation benchmark. Existing benchmarks either overlook personalized information management that is critical for personalization or rely heavily on synthetic dialogues, which exhibit an inherent distribution gap from real-world dialogue. To bridge this gap, we introduce **AlpsBench**, **A**n **L**LM **P**er**S**onalization benchmark derived from real-world human–LLM dialogues. AlpsBench comprises 2,500 long-term interaction sequences curated from WildChat, paired with human-verified structured memories that encapsulate both explicit and implicit personalization signals. We define four pivotal tasks—personalized information *extraction*, *updating*, *retrieval*, and *utilization*—and establish protocols to evaluate the entire lifecycle of memory management. Our benchmarking of frontier LLMs and memory-centric systems reveals that: (i) models struggle to reliably extract latent user traits; (ii) memory updating faces a performance ceiling even in the strongest models; (iii) retrieval accuracy declines sharply in the presence of large distractor pools; and (iv) while explicit memory mechanisms improve recall, they do not inherently guarantee more preference-aligned or emotionally resonant responses. AlpsBench aims to provide a comprehensive framework to accelerate research toward truly personalized AI assistants.

CCS Concepts

• **Information systems** → **Personalization**.

Keywords

LLM Personalization Benchmark; Structured Personalized Memory; Implicit User Preferences

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Table 1: Comparison between AlpsBench and existing LLM personalization benchmarks. "Real" denotes whether the benchmark is built upon real-world data. Columns "T1" to "T4" represent whether the benchmarks contain corresponding testing dimensions.

Benchmark	Real	T1	T2	T3	T4: Utilization Dimensions				
		Ext.	Upd.	Retr.	PA	PF	VRA	CF	EI
LaMP	●	○	○	○	◐	◐	○	○	○
PersonalLLM	○	○	○	○	◐	●	○	○	○
EQ-Bench	○	○	○	○	○	○	○	○	●
PersoBench	○	○	○	○	●	●	○	●	○
PersonaFeedback	○	○	○	○	●	●	○	●	○
LoCoMo	○	◐	○	●	◐	○	○	○	○
LongMemEval	○	●	●	●	◐	○	○	○	○
PersonaLens	○	○	○	◐	●	●	○	●	○
HaluMem	○	●	●	◐	◐	○	○	○	○
PersonaMem v2	○	●	●	◐	●	●	○	○	○
AlpsBench	●	●	●	●	●	●	●	●	●

Category: Memory-free Preference Alignment, Memory-aware Preference Alignment, Ours.
●: Fully Supported / Real Data, ◐: Partially Supported, ○: Not Supported / Synthetic Data.
Ext. = Memory Extraction, **Upd.** = Memory Updating, **Retr.** = Memory Retrieval, **PA** = Persona Awareness, **PF** = Preference Following, **VRA** = Virtual-Reality Awareness. **CF** = Constraints Following. **EI** = Emotional Intelligence.

1 Introduction

Recent advancements in LLMs (*e.g.,* long-context understanding and self-evolution) have demonstrated promising results across different domains [45, 46, 52], opening up significant potential for LLMs to shift from instant general-purpose tools to lifelong evolving AI assistants. Despite that mainstream LLMs are proficient in addressing generic tasks, they still struggle to accommodate heterogeneous users' needs, potentially hurting the user experience in daily use of AI [22, 35]. In the light of this, enabling LLM personalization becomes crucial to achieve lifelong personal intelligence, strongly attracting and motivating both academia [31, 42, 54, 60] and industry (*e.g.,* OpenAI [28], Google [18], and Anthropic [3]) to explore personalizing LLM responses for different individuals.

Facilitating LLM personalization techniques crucially relies on high-quality evaluation benchmarks. Existing benchmarks can be broadly categorized into two groups as shown in Table 1:

- ***Memory-free*** benchmarks (*e.g.,* LaMP [35], personalLLM [65]) focus on the alignment between LLMs' final output and user preference in downstream tasks, such as personalized email generation. Despite the effectiveness in assessing personalization in response, they overlook the process of personalized information governance (*e.g.,* memory extraction, retrieval, update), which is a

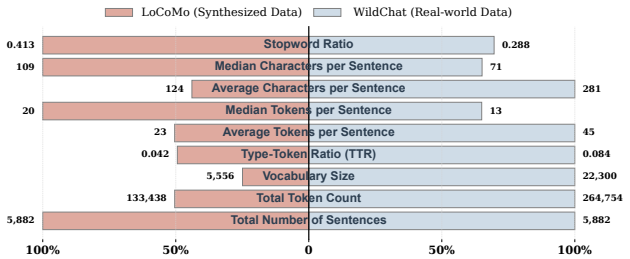

Figure 1: Comparison between synthesized and real data.

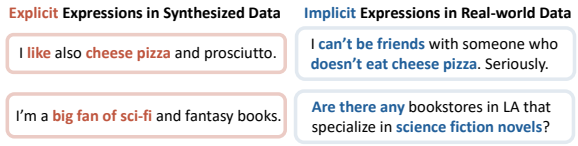

Figure 2: Examples of the synthesized and the real-world human expressions.

key component in understanding and leveraging user preference in personalized responses.

- ***Memory-aware*** benchmarks explicitly incorporate memory evaluation based on synthetic human-LLM dialogues along multiple dimensions (*e.g.,* long-term history memorization [23, 47] and emotional preference alignment [29, 33]). Nonetheless, they suffer from a critical distributional gap between simulated and real-world data, causing two limitations. 1) Lack of diverse conversations, where LLM-synthesized dialogues tend to be homogeneous [14], failing to capture the natural conversational diversity (see Figure 1). 2) Lack of implicit expressions. In real-world interactions, users often convey personalized information implicitly. However, synthetic dialogues can be overly explicit and simplistic (see Figure 2), failing to generalize to real-world scenarios.

In this light, it is imperative to build benchmarks based on real-world user conversational data that capture both conversational diversity and implicit personalization signals. Moreover, instead of simply verifying LLMs' final output, it is essential to systematically evaluate the process of understanding, storing, and utilizing personalized information. Hence, we propose a comprehensive evaluation framework with four core tasks.

- *Task 1: Personalized information* ***extraction***, which evaluates the ability to extract personalized information from dialogues. Given the raw conversational data, the evaluated LLMs are asked to distill them into structured memory containing personalized information (refer to Section 2.1 for details), which is evaluated by F1 score and LLM-as-a-judge against ground-truth memory.
- *Task 2: Personalized information* ***update***, which assesses the capacity to track the dynamics of user information and update structured memory accurately. Given both a historical and a new dialogue, the evaluated LLMs are asked to output updated memory and the memory manipulation (*i.e.,* retention, addition, and modification), assessed by accuracy against ground-truth.
- *Task 3: Personalized information* ***retrieval***, which measures the ability to retrieve relevant personalized information. Given a user query and a candidate set of memory, the evaluated LLMs are asked to select the relevant memory for personalized response. Recall and precision are adopted as evaluation metrics.
- *Task 4: Personalized information* ***utilization***, which examines the alignment of user preference in LLMs' response. We break down the task into measurement of five LLM capabilities, including persona awareness, preference following, virtual-reality awareness, constraint following, and emotional intelligence for a comprehensive alignment of dimensions (refer to Section 2.4 for detailed task designs and evaluation metrics).

To construct a real-world comprehensive benchmark, we develop a four-step pipeline. *Step 1*: Data collection. We leverage a real-world dataset WildChat [58] and collect the dialogues that contains long-term dialogues[1]. *Step 2*: Memory extraction. We design a structured memory taxonomy for LLMs to extract the personalized information from dialogues. *Step 3*: Human Annotation. The extracted memory is then examined by the human annotators to ensure the quality. *Step 4*: Task construction. Based on collected dialogues and the extracted memory, we construct evaluation data for four tasks, including evaluation queries and corresponding ground truth. This pipeline can be continuously updated with evolving real-world human-LLM conversations, enabling a dynamic benchmark.

Building upon the above pipeline, we introduce **AlpsBench**, a comprehensive benchmark for evaluating LLM personalization, with the following key features: 1) AlpsBench consists of 2,500 high-quality real-world human–LLM dialogues from 6 to 249 turns. 2) It covers four tasks for personalized information, *i.e.,* extraction, retrieval, update, and utilization. 3) It assesses multi-dimensional capability of memory utilization, *i.e.,* persona awareness, preference following, virtual-reality awareness, constraint following, and emotional intelligence. We also release a public leaderboard[2] to facilitate continuous benchmarking from the community.

We benchmark frontier LLMs, including general-purpose and memory-oriented LLMs, and drive several key findings: (i) models exhibit limited reliability in extracting latent user traits; (ii) memory updating reaches a performance plateau even for the strongest models; (iii) retrieval accuracy degrades substantially in the presence of large distractor sets; and (iv) explicit memory mechanisms do not inherently ensure stronger preference alignment or emotional resonance in responses. We release our codes at https://github.com/ThisIsCosine/AlpsBench.

The key contributions of this work are summarized as follows:

- We introduce AlpsBench, a dynamic and extensible LLM personalization benchmark built on real-world human–LLM conversations with human-verified structured memory.
- We propose a systematic evaluation framework comprising four core tasks (*i.e.,* personalized information extraction, retrieval, update, and utilization) to benchmark personalization ability.
- Extensive evaluations on AlpsBench reveal that existing models remain limited in reliable long-term personalization and context-aware utilization, highlighting substantial opportunities for future research in personalized LLMs.

[1]WildChat is licensed under ODC-BY (Open Data Commons Attribution). In accordance with this license (including the attribution requirement), we release our reprocessed version under the same ODC-BY license, intended for academic and research purposes.
[2]https://misshsiaoo.github.io/Alps_Bench/.

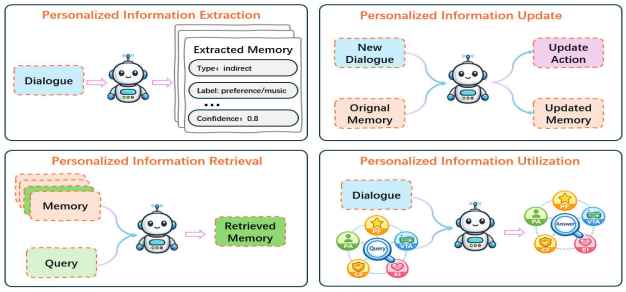

Figure 3: The evaluation tasks of AlpsBench.

2 AlpsBench Evaluation Framework

To rigorously evaluate the personalization capabilities of AI assistants, we introduce AlpsBench, a comprehensive benchmark designed to simulate real-world personalization scenarios. In this section, we provide a detailed description of the task design for AlpsBench (*rf.* Figure 3), which comprises four core tasks specifically designed to assess different facets of personalized AI assistants.

2.1 Task1: Personalized Information Extraction

A core capability of personalized AI assistants lies in their proficiency in transforming raw conversational data into structured, high-quality memories. This capability reflects the assistants' depth of understanding regarding historical user preferences and key information. Motivated by the need to assess this capability, we propose a *Personalized Information Extraction* task:

$$\mathcal{H} \xrightarrow{\text{AI assistant}} [\hat{\mathcal{M}}_1, \hat{\mathcal{M}}_2, \ldots], \tag{1}$$

where the AI assistant is tasked with extracting structured user information from the original user dialogue history ($\mathcal{H}$). The resulting output consists of a set of memories ($\hat{\mathcal{M}}_1, \hat{\mathcal{M}}_2, \ldots$), each containing the following attributes:

- **Memory ID:** A unique identifier for the memory entry.
- **Memory Type:** Explicit or implicit preference.
- **Label:** The taxonomic category of the memory.
- **Value:** The exact statement extracted from the original dialogue.
- **Confidence:** The confidence level of the extraction.

The quality of the extractions (*i.e.,* memories) is evaluated by comparing them against human-annotated ground truth ($\mathcal{M}$). Specifically, we calculate the F1 score for exact matches, measuring both the coverage and quality of the memories relative to the ground truth. Furthermore, we leverage an LLM to assess semantic similarity, ensuring a more nuanced evaluation beyond lexical matching.

2.2 Task2: Personalized Information Update

The dynamic evolution of user preferences requires personalized AI assistants to possess a sophisticated memory-updating mechanism. This process hinges on three fundamental capabilities: (i) the robustness in filtering dialogue noise, (ii) the ability to add new user preferences, and (iii) the logic for resolving preference conflicts by reasonably updating outdated memories with new information. Thus, we introduce the *Personalized Information Update* task:

$$(\mathcal{M}, \mathcal{H}_{new}) \xrightarrow{\text{AI assistant}} (\hat{\mathcal{M}}_{new}, \hat{\text{Act}}), \tag{2}$$

where the task requires assistant to output the updated memories $\hat{\mathcal{M}}_{new}$ and categorize each update according to an action type $\hat{\text{Act}} \in$ [Retention, Addition, Modification], corresponding to the aforementioned capabilities. The evaluation of this task involves calculating the accuracy of the predicted actions against the human-annotated ground truth. Additionally, the updated memories can be leveraged for further analysis of the AI assistants (*c.f.,* subsection 4.1.2).

2.3 Task3: Personalized Information Retrieval

The ability to retrieve relevant memories is a cornerstone of an AI assistant's personalization. Accurately identifying useful insights from a user's vast memory pool significantly enhances the AI assistant's capacity to provide tailored responses. In this light, we introduce the *Personalized Information Retrieval* task:

$$(\mathcal{M}_{pos}, [\mathcal{M}_{neg1}, \ldots.], \mathcal{Q}) \xrightarrow{\text{AI assistant}} \hat{\mathcal{M}}_{pos}, \tag{3}$$

where the AI assistant is expected to retrieve the most relevant memory ($\hat{\mathcal{M}}_{pos}$) based on the user query ($\mathcal{Q}$) from a set of candidate memories, which consists of one positive sample ($\mathcal{M}_{pos}$) labeled by humans and several randomly sampled negative samples ([$\mathcal{M}_{neg1}$,....]). We use the widely recognized metric of recall to report the evaluation results of this task.

2.4 Task4: Personalized Information Utilization

The ability to utilize users' personal information to address real-world needs is fundamental to personalized AI assistants. To systematically evaluate this capability, we introduce the *Personalized Information Utilization* task, formulated as follows:

$$(\mathcal{H}, \mathcal{Q}) \xrightarrow{\text{AI assistant}} \hat{\mathcal{R}}, \tag{4}$$

where the assistant is tasked to produce a response $\hat{\mathcal{R}}$ to a user query $\mathcal{Q}$ based on dialog history $\mathcal{H}$.

Based on the characteristics of users' real-world needs, we design five core evaluation dimensions for this task:

Persona Awareness (PA). This dimension assesses whether the assistant correctly recalls and applies explicit user attributes, such as educational background. Users often provide explicit persona information across multiple sessions, requiring the AI assistant to seamlessly integrate this data into its responses.

Preference Following (PF). This dimension measures whether the assistant can capture implicit user preferences within dialogue history and align its responses accordingly. Beyond explicit persona information recall, it tests inductive reasoning to align responses with potential tastes and habits—a critical capability for personalized recommendation and planning.

Virtual-Reality Awareness (VRA). This dimension evaluates whether the assistant can distinguish real user information from role-play or fictional content, ensuring that in-character data does not contaminate real-world assistance. For example, if a middle-aged professional previously role-played as a student, the assistant should disregard student's attributes when drafting his/her resume.

Constraint Following (CF). This dimension examines whether the assistant respects constraints expressed by the user in prior interactions. There may be instances where the user explicitly requests that certain information be excluded or specific conditions

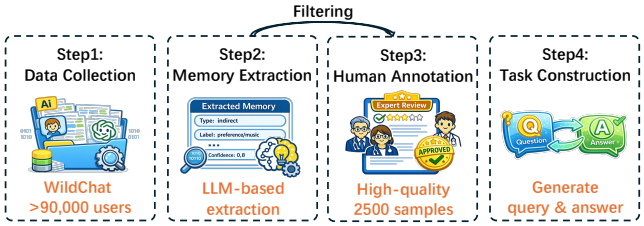

Figure 4: The four-step benchmark construction pipeline based on real human–LLM dialogues

be met. The assistant should consistently follow these guidelines to ensure alignment with the user's expectations.

Emotional Intelligence (EI). This dimension assesses whether the assistant uses user dialogue history to provide emotionally appropriate responses. An effective AI assistant enables differentiated emotional responses, such as offering encouragement to resilient users and reassurance to more sensitive users.

In this task, we adopt LLM-as-a-Judge to evaluate AI assistants. For each dimension (except for Emotional Intelligence), the judge determines whether the AI assistant's response meets the specified criteria and assigns a binary score (0 or 1) accordingly. Since Emotional Intelligence requires a more nuanced evaluation, we follow the evaluation framework of [44], which uses the LLM as a reward model to assign a score on a 1–5 scale.

3 Curation Pipeline for AlpsBench

In this section, we describe the four-step curation pipeline of AlpsBench, as depicted in Figure 4.

- **Step 1: Data Collection**. Personalized AI assistants are intended to serve actual human needs. Therefore, to effectively test their capabilities, the design of the personalization benchmark should be grounded in the real-world human-AI interactions. To achieve this, we use the WildChat [58] dataset, which contains approximately 90,000 users' interaction records covering a broad range of authentic conversational scenarios. Considering longer dialogues inherently carry richer contextual dependencies and cross-temporal information, we further refined the data by selecting dialogues with more interaction turns. Each dialogue in this subset contains between 6 and 249 turns of interaction.
- **Step 2: Memory Extraction and Filtering**. To enable a systematic evaluation of an AI assistant's ability to process and utilize personalized information, we first employ the *DeepSeek-v3.2 reasoning model* [6] to automatically extract structured information from raw user dialogue histories and format it into memory representations, as illustrated in Section 2.1.

Considering the substantial computational cost of running AI assistants and the expense of high-quality manual annotation (*cf.* Step 3), we further perform data filtering based on high-level semantic categories in the structured memories (*i.e.,* the *label* attribute). Specifically, we subsample instances within each category, ensuring that the number of samples per category does not exceed a predefined upper bound. This strategy effectively removes large amounts of semantically redundant content.

Moreover, to enhance the challenge and discriminative power of the benchmark, we prioritize retaining users whose memories contain implicit information while performing category-based filtering. Specifically, we retain a user u if there exists at least one memory $\mathcal{M}$ with the type attribute labeled as *implicit*. Formally:

$$U_{\text{retained}} = \{u \in U \mid \exists \mathcal{M} \in \mathcal{M}_u : \text{type}(\mathcal{M}) = \textit{implicit}\}, \tag{5}$$

where U is the set of all users. $\mathcal{M}_u$ is the set of memories associated with user u. Ultimately, this process yields 2,500 user samples that constitute the final benchmark.

- **Step 3: Human Annotation**. To ensure the reliability of the benchmark, we further perform expert-level manual verification of the automatically extracted memories. Specifically, we invite senior practitioners with over one year of relevant domain experience to review and correct the structured memories. This rigorous vetting process establishes a high-quality ground truth, enabling a robust evaluation for personalized AI assistants.
- **Step 4: Task Construction**. Based on the collected dialogue histories and human-verified memories, we construct evaluation queries and corresponding ground truth for the four tasks described in Section 2. Selecting a memory as a target memory ($\mathcal{M}_{tgt}$) and using the original user dialogue ($\mathcal{H}$) as context, we employ *GPT-5.2* [27] to synthesize task-specific queries and ground-truth answers. This process can be formulated as:

$$(\mathcal{H}, \mathcal{M}_{tgt}, [*]) \xrightarrow{\text{GPT-5.2}} (\hat{\mathcal{Q}}, \hat{\mathcal{A}}) \xrightarrow{\text{Human}} (\mathcal{Q}, \mathcal{A}) \tag{6}$$

where $[*]$ represents task-specific auxiliary inputs. In the case of ***Task2***, its $[*]$ represents target update strategies—specifically Retention, Addition, and Modification—to guide the generation of update-oriented queries. In the ***Task3***, its $[*]$ represents randomly selected memories as distractors (negative samples) to enhance the realism and complexity of the retrieval tasks.

The output of this process consists of two main elements: $\mathcal{Q}$ and $\mathcal{A}$. Specifically, for ***Task1***, $\hat{\mathcal{Q}}$ consists of system instructions for memory extraction, while $\hat{\mathcal{A}}$ represents the $\mathcal{M}_{tgt}$ itself. For ***Task2***, it $\hat{\mathcal{Q}}$ corresponds to the user's new dialogue history along with the system's instruction (*e.g.*, "Please update memory") and $\hat{\mathcal{A}}$ reflects the applied update strategy. For ***Task3***, $\hat{\mathcal{Q}}$ is a practical user inquiry related to the $\mathcal{M}_{tgt}$, with $\hat{\mathcal{A}}$ as the $\mathcal{M}_{tgt}$. Finally, for ***Task4***, $\hat{\mathcal{Q}}$ is a user query related to the $\mathcal{M}_{tgt}$topic, and there $\hat{\mathcal{A}}$ is the generated response addressing that query.

Notably, all generated pairs $(\hat{\mathcal{Q}}, \hat{\mathcal{A}})$ undergo rigorous quality control by human ensuring that the queries contain no information leakage regarding the $\mathcal{M}_{tgt}$ and that all answers are factually accurate. The resulting verified pairs $(\mathcal{Q}, \mathcal{A})$ constitute the final benchmark for evaluation.

4 Experiments

- **Evaluated Models**. For **general-purpose LLMs**, we include **open-source models** (DeepSeek-v3.2 in thinking mode [6], Qwen3-max [51], Llama-4 Maverick [24]) and **closed-source models** (GPT-5.2 [27], GPT-4.1-mini [26], Gemini-3-Flash-preview [8], Claude-Sonnet-4.5 -2025032 [2]). For **memory-oriented systems**, we implemented MemoryOS [15], EverMemOS [10], A-Mem [50], LightMem [7], MemOS [19], Mem0 [5], and Mem0g [5]. All memory systems use **GPT-4.1-mini** as the backbone LLM. For Task 3 in the general-model setting, we additionally evaluate two standard retrieval baselines for memory models, **nltk [41] +bm25 [32]** and

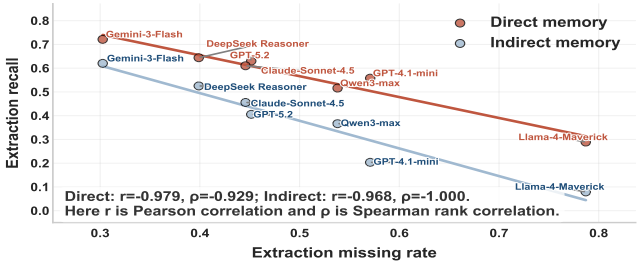

Figure 5: Task 1. Comparison of missing rate and recall across direct vs. indirect personalized memory.

all-MiniLM-L6-v2 [37]. The reason is these classic retrieval methods are widely used as external retrievers for memory retrieval.

- **Implementation Details**. We use GPT-5.2 [27] as the unified generator for query generation, and DeepSeek-v3.2 [6] as the judge model for automatic evaluation.

4.1 Performance on AlpsBench

4.1.1 Task 1: Extraction Result Analysis. **General-purpose LLMs remain limited in personalized information extraction.** As shown in Table 2, even the best-performing model, Gemini-3 Flash, achieves only 51.67 on Extraction, while Llama-4 scores as low as 22.07, raising concerns about the robustness of current models for practical deployment. Moreover, these models perform substantially better when personalized information is conveyed through **direct expressions** than through **indirect expressions** (Figure 5), suggesting that extracting personalized information embedded in indirect expression remains challenging. To understand the source of errors, we find that the **extraction missing rate** is most strongly associated with performance degradation, showing a strong negative correlation with indirect-memory extraction recall (Spearman $\rho = -1.000$, Pearson $r = -0.959$). In contrast, correlations with hallucination and redundancy are weaker and less stable (Spearman $\rho = -0.464$ and 0.250, respectively); due to page limitations, the full statistical analysis is available in our Git repository. **Turning to memory-oriented systems,** Figure 6 further shows that they can underperform their backbone in extraction recall. One possible explanation is the **storage-policy bias**: these systems may retain overly broad dialogue content (i.e., "store everything") rather than prioritizing personalization-relevant information.

4.1.2 Task 2: Update Result Analysis. **General-purpose LLMs exhibit a noticeable performance ceiling in the personalized information updating task.** According to Table 2, although GPT-5.2 delivers the best performance overall, its score of 81.49 still falls short of a truly satisfactory level. Notably, open-source models **DeepSeek Reasoner** (80.91) and **Qwen3-max** (76.28) follow closely behind the leader, demonstrating that frontier open-source models have attained a level of proficiency comparable to proprietary systems in handling complex memory updating requirements. The overall performance of general-purpose models suggests that, across both model categories, current LLMs remain limited in their ability to integrate new user updates into persistent personalization memory. Moreover, as shown in Figure 7, **general-purpose LLMs' memory updating performance on new-memory addition and conflict-memory modification shows a weak association.**

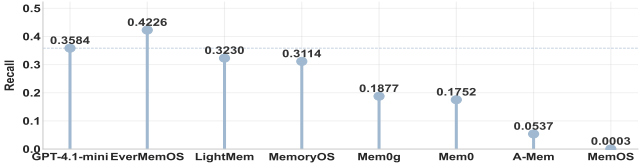

Figure 6: Task 1. Extraction recall of memory-oriented systems compared with backbone model (GPT-4.1-mini)

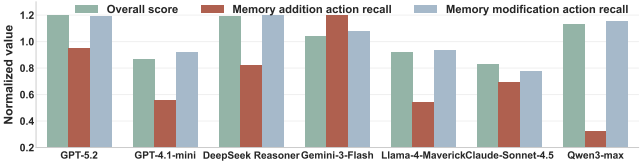

Figure 7: Task 2. Updating performance of general-purpose LLMs: overall, memory addition and modification scores.

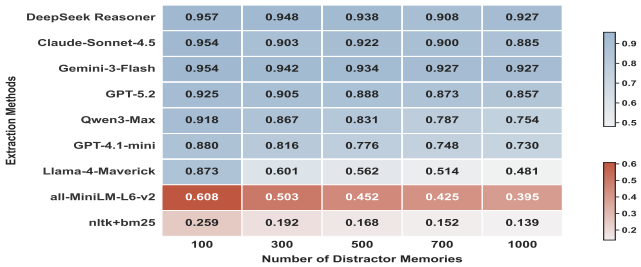

Figure 8: Task 3. Retrieval performance decay across evaluated models as distractor memory count increases.

While the green bars summarize overall Task 2 performance, the red (new-memory addition) and blue (conflict-memory modification) bars exhibit noticeably different patterns across models, suggesting limited consistency. This indicates that two forms of **memory update actions**—appending previously unseen personalized information to the stored memories versus **editing/overwriting existing memories** when new dialogue introduces conflicts—may rely on distinct mechanisms: strong performance on one does not guarantee reliable performance on the other, and vice versa.

4.1.3 Task 3: Retrieval Result Analysis. **The sensitivity of general-purpose LLMs to distractor memories varies significantly, with the performance gap between models expanding as distractor density increases.** As evidenced by **Figure 8**, all evaluated models maintain high and relatively uniform retrieval performance in low-noise environments (100 distractors), with scores tightly clustered between 0.8729 and 0.9569. In high-noise settings (1000 distractors), SOTA LLMs remain robust: Gemini-3 Flash and DeepSeek Reasoner both achieve scores above 0.92, demonstrating strong resilience in retrieval tasks. However, given the inherent limitations of context window size and the escalating token costs associated with long-context processing, improving performance in realistic, long-horizon multi-turn scenarios requires a more lightweight and efficient retrieval architecture. **Also, current semantic-level retrieval methods are largely inadequate for personalized memory.** There is a stark performance disparity between frontier LLMs and the retrieval components typically integrated into memory frameworks. Classical statistical methods such as *nlk+BM25* and semantic embedding baselines like *all-MiniLM-L6-v2* experience a

Table 2: Tasks 1–4. Experimental evaluation result of general-purpose LLMs.

Model	Task 1 Extraction	Task 2 Update	Task 3 Retrieval					Task 4 Utilization						
			Number of Distractor Memories					PA	PF		VRA	CF	EI	
			100	300	500	700	1000		Gen.	Int.			EN	CN
GPT-5.2	41.43	**81.49**	0.9254	0.9052	0.8884	0.8733	0.8572	0.5702	0.6983	**0.7680**	0.5702	0.5702	3.42	3.90
GPT-4.1-mini	33.69	54.66	0.8802	0.8156	0.7761	0.7482	0.7295	0.4018	0.4995	0.5240	0.4018	0.4018	2.79	2.92
DeepSeek Reasoner	47.79	80.91	**0.9569**	**0.9484**	**0.9376**	0.9083	**0.9273**	0.5825	0.6483	0.6120	0.5825	**0.9602**	3.66	**4.00**
Gemini-3 Flash	**51.67**	68.85	0.9538	0.9419	0.9342	**0.9268**	0.9269	**0.6895**	**0.7655**	0.7052	**0.6895**	0.8328	3.49	3.58
Llama-4 Maverick	22.07	58.84	0.8729	0.6005	0.5616	0.5141	0.4811	0.2684	0.1152	0.3080	0.1552	0.8720	2.48	2.38
Claude-Sonnet-4.5	41.64	51.25	0.9542	0.9030	0.9222	0.8999	0.8855	0.5614	0.6045	0.5498	0.5933	0.9514	3.10	3.05
Qwen3-max	39.01	76.28	0.9180	0.8669	0.8314	0.7871	0.7542	0.6228	0.6901	0.6574	0.6834	0.8267	**3.68**	3.84

PA = Persona Awareness, **PF** = Preference Following, **VRA** = Virtual-Reality Awareness, **CF** = Constraint Following, **EI** = Emotional Intelligence.

Table 3: Task 4. Utilization performance comparison of memory-oriented systems.

Method	PA	PF		VRA	CF	EI	
		Gen.	Int.			EN	CN
GPT-4.1-mini	0.4018	0.4995	0.5240	0.2780	0.7774	2.79	2.87
Grd. Mem.	-	-	-	-	-	3.21	3.33
MemoryOS	0.3719	0.5117	0.5458	0.1500	0.7660	2.23	2.26
A-Mem	0.2825	0.5566	0.4303	**0.2460**	**0.8532**	1.98	1.95
EverMemOS	**0.7246**	**0.8104**	**0.6494**	0.1580	0.7356	**2.68**	**2.73**
Mem0	0.2719	0.4557	0.3825	0.1500	0.8110	1.88	2.02
Mem0g	0.2754	0.4516	0.3904	0.1580	0.8171	1.87	1.91
MemOS	0.2632	0.5423	0.4502	0.1300	0.8323	2.05	2.08
LightMem	0.2438	0.4455	0.4025	0.1854	0.8208	1.79	1.83

Grd. Mem. denotes presenting the LLM the query along with the ground truth memories.

total performance collapse as distractor density scales, with accuracy plummeting to 0.14 and 0.40, respectively. This suggests that "plug-and-play" retrieval modules are fundamentally inadequate for the complexities of long-term personalization. Future research should prioritise the development of logic-aware retrieval layers capable of bringing LLM-level reasoning to the memory extraction process without the prohibitive costs of full-context processing.

4.1.4 Task 4: Utilization Result Analysis. **Existing LLMs struggle to perform well across all dimensions of personalized memory utilization.** From the results shown in Table 2, no single general-purpose LLM consistently dominates across evaluation dimensions. While Gemini-3 Flash performs best on Persona Awareness, DeepSeek achieves the strongest performance on Constraint Following, suggesting that personalized agentic competence remains fragmented across models. Moreover, even SOTA LLMs exhibit limited absolute performance—for instance, Gemini-3 Flash attains only 0.6895 on Persona Awareness, which emphasizes the need for memory-augmented personalized systems.

While memory systems enhance model capabilities, they also introduce "personalization bias." From the results shown in Table 3, we observe that recent memory systems improve certain aspects of persona utilization. For example, EverMemOS improves GPT-4.1-mini by over 80% on both Persona Awareness and Persona Preference. However, these gains come with notable drawbacks in Virtual-Reality Awareness and Emotional Intelligence. For instance, even the best-performing memory-augmented system achieves only 2.68 on English, falling short of the backbone's 2.79. We argue that these limitations stem from current design strategies that prioritize memory capacity over effective memory usage. Existing systems emphasize recording and retrieving information, while largely overlooking the filtering and verification processes required for reliable memory formation. Moreover, their problem-solving–oriented design biases responses toward task resolution, often at the expense of emotional regulation and high-EQ interaction. To enable reliable real-world deployment and deliver emotionally meaningful user experiences, future memory systems should move beyond merely expanding memory capacity and instead prioritize memory accuracy, robust filtering, and the effective utilization of stored information to generate responses aligned with users' preferences.

5 Related Work

LLM Personalization. LLM personalization research [20, 38, 43, 54, 61] primarily unfolds along three dimensions [9, 21, 30, 48, 55, 56, 59]. Personalized Prompting retrieves user-specific context without modifying parameters, evolving from simple storage to agentic architectures: Mem0 [5] and MemInsight [34] establish persistent memory layers; A-MEM [50] and Nemori [25] introduce autonomous decision-making; LightMem [7] employs a cognitive "Sensory-Short-Long" architecture with sleep-time updates; while MemOS [19] and MemoryOS [15] elevate memory to system-level resources, and EverMemOS [10] utilizes Engram-inspired lifecycle management. Personalized Adaptation focuses on parameter-efficient fine-tuning to internalize user patterns. Moving beyond PLoRA [53]'s specific adapters, RecLoRA [64] and iLoRA [16] leverage Mixture-of-Experts with dynamic routing for shifting preferences, while OPPU [39] balances privacy via collaborative training. Finally, Personalized Alignment optimizes training objectives: MORLHF [49] improves RL via multi-objective rewards, MODPO [63] integrates alignment into DPO, and Personalized Soups [11] merges parameters at inference for diverse preferences.

LLM Personalization Benchmarks. Existing benchmarks can be broadly categorized into two groups: Memory-free and Memory-aware. The first category, represented by LaMP [35], LongLaMP [17], and LaMP-QA [36], focus on aligning generated content with user preferences in downstream tasks via retrieved static history. Similarly, PersonalLLM [65], PersonaFeedback [40], and PersoBench [1] evaluate adherence to pre-defined profiles. However, by treating user history as fixed context, these methods overlook the critical process of personalized information governance (*e.g.,* memory extraction, retrieval, and update). In contrast, Memory-aware benchmarks incorporate multi-dimensional assessments of memory capabilities. Early works like EmoBench [33] and EQ-Bench [29] targeted emotional alignment, while LoCoMo [23] and LongMemEval [47] long-term memorization. Specific diagnostics such as PrefEval [57]

and HaluMem [4] further address preference consistency and memory hallucinations. Recently, agentic simulations like PersonaLens [62], PersonaMem [12], and its successor PersonaMem v2 [13] have been proposed to track profile evolution. Despite these advances, their reliance on synthetic human-LLM dialogues creates a significant distributional gap from real-world data, resulting in homogeneous conversations [14] that lack natural diversity and fail to capture the implicit expressions characteristic of authentic human interactions.

6 Conclusion and Future Work

In this paper, we presented AlpsBench, a benchmark designed to evaluate the complete lifecycle of LLM personalization using real-world dialogue data. By leveraging long-term human-LLM interactions and expert-verified structured memories, AlpsBench enables a granular diagnosis of LLM performance across four key dimensions: personalized information extraction, update, retrieval, and utilization. Extensive experiments highlight several critical challenges for current LLMs, including difficulties in interpreting implicit user information, handling preference drift, and maintaining retrieval reliability under heavy interference. Going forward, we will continuously maintain and update the dataset, making it a dynamic benchmark to evaluate emerging LLMs.

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